HARSHAL RAAJ PATNAIK

Senior Software Engineer | Cloud Platforms, Distributed Systems & Technical Lead (Tech × MBA)

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EDUCATION

Indian Institute of Management Kozhikode (IIMK) **2025 – Present**

Executive Post Graduate Programme (EPGP) in General Management – Product Strategy, Finance & Operations *Kochi, India*

SRM Institute of Science and Technology **2017 – 2021**

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.76 / 87.6%) *Chennai, India*

PROFESSIONAL EXPERIENCE

Novamesh Ltd. (Wholly Owned Subsidiary of Tata Communications) **Apr 2024 – Present**

Senior Software Engineer (Cloud Platforms & Distributed Infrastructure) *Chennai, India*

- Architected multi-hypervisor provisioning and capacity reconciliation engine across **KVM, VMware ESXi, and OLVM**; decoupled long-running provisioning tasks via asynchronous **RabbitMQ** queues to guarantee sub-second portal responsiveness.
- Slashed cloud resource batch scheduler latency by **92.6% (from ~18 minutes down to 80 seconds)** across distributed Cassandra clusters by implementing strict 50-item batch chunking, multi-level Redis caching, and tuning HikariCP connection pools.
- Automated network security controls via **FortiGate REST APIs** for firewall rules, VDOM tokens, and IPsec VPNs; engineered an in-process sliding-window token bucket rate limiter that **eliminated 100% of HTTP 429 rate-limiting failures**.
- Orchestrated multi-step asynchronous infrastructure workflows using the **Flowable BPMN engine**; designed automated compensating sagas to guarantee zero orphaned cloud resources during hypervisor provisioning failures.
- Engineered route discovery, wildcard forwarding, and multi-tenant access control across **Kong API Gateway**; stepped up as **Acting Scrum Master** and technical mentor, driving architecture reviews, code quality, and on-time sprint deliveries.

Tata Communications Ltd. **Jul 2021 – Mar 2024**

Software Development Engineer (Cloud Infrastructure, Storage & Full-Stack) *Chennai, India*

- Built full-stack onboarding workflow for **Caringo Swarm S3-compatible cloud object storage** and automated cPanel DNS A-record associations using Java and Angular, enabling **100% self-service storage provisioning and bucket metering**.
- Spearheaded enterprise frontend modernization to **Angular 17** with reactive component architectures, deconstructing legacy AngularJS scopes to achieve a **35% improvement in client-side rendering speed** and state predictability.
- Automated continuous integration and release packaging pipelines with **Jenkins CI/CD**, multi-module Maven builds, and automated JUnit regression test suites, cutting release turnaround time from days to automated hours.
- Engineered switch-port utilization and network capacity telemetry engine integrating Zabbix and Nimsoft APIs, computing cluster-wide compute and datastore allocation to prevent capacity exhaustion; earned **3× Extra Mile Commitment Awards**.
- Integrated multi-vendor load balancer orchestration across **HAProxy, F5 BIG-IP, and VMware Avi** with automated VIP provisioning, deep application health probes, and tenant-scoped SSL/TLS offloading.

STRATEGY & TECHNICAL PROJECTS

SthiraX: Distributed CO_2-Mineralization & IoT MRV Platform | *IIM Kozhikode Capstone (BPP)* **2025 – Present**

- Designed an asset-light retrofit architecture converting regional precast concrete plants into certified low-carbon facilities to fulfill India's **April 2026 C&D Waste Rules (mandating 5% to 25% recycled material by 2030 across 20,000+ sqm projects)**.
- Architected batch-level **IoT MRV (Measurement, Reporting & Verification) software** generating cryptographically verifiable Digital Product Passports for developer EPR compliance, supported by multi-tiered SaaS subscriptions and per-tonne royalty models.

Workflow Reliability Lab: Distributed Sagas & Chaos Engineering | *Distributed Systems Platform* **2024 – 2025**

- Engineered a developer platform for testing and verifying distributed workflow reliability; built synthetic fault injection simulators (network partitions, timeout anomalies) and invariant checkers verifying ACID saga compensations (**Java 25, Spring Boot, Angular, Docker**).

QueryLens AI & CloudScale: Sandboxed Text-to-SQL & Predictive Autoscaler | *AI & Cloud Systems* **2024 – 2025**

- Architected a modular C4 3-tier Text-to-SQL engine (Angular → Spring Boot → FastAPI with pgvector RAG) with read-only sandbox roles preventing SQL injection; paired with predictive ML capacity autoscaling and Prometheus/Grafana telemetry.

TECHNICAL & LEADERSHIP COMPETENCIES

Languages & Core: Java (8/11/17/25 LTS), Python, TypeScript, JavaScript, SQL, HTML5/CSS3, Bash
Distributed Systems & Cloud: Spring Boot 3/4, Flowable BPMN, RabbitMQ (AMQP), Microservices, REST APIs, KVM, VMware ESXi, OLVM, Docker
Databases, Caching & Storage: Cassandra, Redis, MySQL, PostgreSQL (pgvector), Caringo Swarm / S3, Hibernate/JPA, HikariCP Tuning
Networking, Security & Gateways: Kong API Gateway, FortiGate REST APIs, IPsec VPN, DNS Automation, Sliding-Window Token Bucket, Reverse Proxies
DevOps, SRE & Observability: Jenkins CI/CD, Prometheus, Grafana, OpenTelemetry, JUnit, Mockito, Zabbix APIs, Linux, Maven, SonarQube
Product Strategy & MBA Coursework: Product Management, Financial Accounting, Operations Management, Strategy Management, Risk Analysis, Product Analytics, Negotiation & Conflict Management, Entrepreneurship

HONORS, CERTIFICATIONS & EXTRACURRICULAR

- **3× Extra Mile Commitment Award** – Tata Communications Ltd. for critical cloud platform delivery, stability, and zero-defect deployments.
- **Flipkart Grid 2.0 Engineering Challenge** – Team Lead; achieved **Top 150 Nationwide Rank** out of 20,000+ teams (Computer Vision & OCR).
- **Architecting with Google Compute Engine** (Google Cloud / Coursera) & **Full Stack Web Development** (Coding Ninjas).
- **Rank 2404 Globally** in Second Round of 5th SOF International Mathematics Olympiad (Science Olympiad Foundation).